

Training and evaluating the model

Nested cross-validation for CatBoost · 187 patients · random seed 42

■ used to fit

■ tuning validation

■ test set (held out)

■ search / decision

Which patients are held out

Repeated Stratified K-Fold: 4 folds × 10 repeats = 40 train/test splits

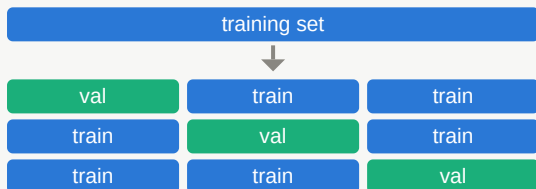


×10

repeats

Which training patients tune the model

Stratified K-Fold: 3-fold, training set only



Optuna TPE — 100 trials per train/test split

Objective: mean AUPRC across the 3 tuning folds

Tunes: depth · learning rate · L2 leaf reg · pos-class weight

carried forward: winning hyperparameters + the same 3 tuning folds

Refit, threshold, and test-set evaluation

· per split

1 · Refit

Winning params refit on the full training set

2 · OOF scores

One score per patient

3 · Threshold

Maximize F1 on OOF curve

4 · Test-set evaluation

Model + threshold scored once
→ performance metrics and plots